

心脏病预测的GDBT模型超参数调优

吕庆聪 山东工商学院 计算机科学与技术系 中国烟台 qclv@sdtbu.edu.cn

摘要

根据世界卫生组织的数据，心脏病在过去二十年里一直是死亡的主要原因。心脏病患者会出现心悸、胸痛、呼吸困难等症状。如何根据这些症状诊断心脏病是医生和科学家的一个重要研究课题。利用机器学习对心脏病进行分类被认为是一种可靠且准确的心脏病检测技术。在本研究中，我们旨在找到一个良好的模型来预测心脏病。为了找到最佳模型，我们比较了三种GBDT（梯度提升决策树）模型，即XGBoost、LightGBM和CatBoost，以识别人是否患有心脏病。我们通过调整超参数来比较三种模型，以提高心脏病预测的准确性。为了对比结果，我们建立了三个使用默认超参数的基准模型。研究结果显示，经过超参数调整的CatBoost模型是最适合的心脏病预测模型。

关键词-心脏病；XGBoost；LightGBM；CatBoost；超参数调优

1 引言

根据数据显示，心脏病已成为危害人类健康的头号杀手。每年约有1750万人死于心脏病，占全球死亡总数的30%。心脏病严重影响人们的生活和健康。心脏病患者会出现心悸、胸痛、呼吸困难等症状。如何根据这些症状诊断心脏病是医生和科学家们的重要研究课题。利用机器学习对心脏病进行分类被认为是一种可靠且准确的心脏病检测技术。在本研究中，我们使用了三种梯度提升决策树（XGBoost、LightGBM和CATBoost）根据心脏病特征进行分类。首先，我们对数据进行了预处理，包括重复值处理、缺失值处理和异常值处理。我们对数据进行的初步分析，包括特征之间的相关性、性别、年龄、血压等特征之间的关系

本文其余部分的组织安排如下。第二部分回顾了心脏疾病预测的相关工作。第三部分介绍了研究方法、数据预处理方法、超参数调优技术以及三个分类器。第四部分比较了三个分类器模型在默认参数和可调超参数下的结果。最后一部分对本文进行总结并讨论未来的工作。

2 相关工作

已经进行了大量研究，使用机器学习方法来预测心脏病。各种机器学习方法的不同实现，如支持向量机、逻辑回归、贝叶斯信念网络等，已经被应用。Anirban Ghosh 使用五种不同的分类模型（逻辑回归、支持向量机、随机森林模型、朴素贝叶斯分类器和线性判别分析）来预测心脏病。作者认为 RFM 是预测心脏病最合适的指标。

Hyperparameter Tuning of GDBT Models for Prediction of Heart Disease

Lv Qingcong
Department of Computer Science and Technology
Shandong Technology and Business University
Yantai, China
qclv@sdtbu.edu.cn

ABSTRACT

According to World Health Organization, heart disease has remained the leading cause of death for the last twenty years. Patients with heart disease will have palpitations, chest pain, dyspnea and other characteristics. How to diagnose heart disease based on these characteristics is an important research topic for doctors and scientists. Classifying heart disease with machine learning is known as a reliable and accurate heart disease detection technique. In this study, we aim to find a good model for predicting heart disease. To find the best model, we compare three GBDT(Gradient Boosting Decision Tree) models i.e., XGBoost, LightGBM and CatBoost to identify whether a person has heart disease or not. We compare the three models tuning the hyperparameters to improve heart disease predicting. To compare results, we create three base models with default hyperparameters. The findings demonstrate that the CatBoost with hyperparamer tuning model as the the most appropriate heart disease prediction model with accuarcy of 90%. Results indicated that tuning the hyperparameter improved modeling results.

Keywords-Heartdisease;XGBoost;LightGBM;CatBoost;Hyperparameter tuning

1 INTRODUCTION

According to data, heart disease has become the number one killer of human health. Every year about 17.5 million people die of heart disease, accounting for 30% of all deaths worldwide. Heart disease seriously affects people's life and health. Patients with heart disease will have palpitations, chestpain, dyspnea and other characteristics. How to diagnose heart disease based on theses characteristics is an important research topic for doctors and scientists. Classifying heart disease with machine learning is known as a reliable and accurate heart disease detection technique. In this study, we used three Gradient Boosting Decision Trees(XGBoost, LightGBM and CATBoost) to classify heart disease based on its characteristics. First, we preprocess the data, including repeated value processing, missing value processing and outlier processing. Our preliminary analysis of the data, including the correlation between features, the relationship between gender, age, blood pressure and other features and the disease. Second, we build three Gradient Boosting Decision Trees for classification prediction of heart disease. The three Gradient Boosting Decision Trees are XGBoost, LightGBM and CATBoost. Third, for each classifier, we conduct two types of experiments with default hyperparameters and adjustable hyperparameters, and compare their Accuracy, Recall, Roc_Auc and Precision metrics results. The experimental results show that using hyperparameter tuning will bring about a significant improvement in performance indicators.

The remainder of this paper is organized as follows. Related work in heart disease prection is reviewd in Section II. In Section III, we describe the research methodology, data preprocessing method, hyperparameter tuning technique, three classifiers. Section IV compares the results of three classifier models with default parameters and adjustable hyperparameters. The last section concludes the paper and discusses future work.

2 RELATED WORKS

Lot of research has been carried out to predict heart disease using machine learning methods. Different implementation of machine learning methods such as Support Vector Machine, Logistic Regression, Bayesian Belief Network, etc. have been applied. Anirban Ghosh using five different classification models (Logistic Regression, Support Vector Machine, Random Forest Model, Naive Bayes classifier and Linear Discriminant analysis) to predict heart disease. The author thinks RFM is the most appropriate predictor of heart disease^[1].

分类器通常需要大量的数据特征进行训练。为了识别对分类结果贡献最大的特征，作者使用Shapley值作为分析心脏病的辅助工具。为了对心脏病数据进行分类，作者使用包含20棵决策树的随机森林分类器。Pal等人也使用随机森林模型来预测心脏病。他们获得了86.9%的准确率，90.6%的灵敏度和82.7%的特异性。

Polipireddy Srinivas 和 Rahul Katarya 提出了一个新模型 hyOPTXg 来识别心血管疾病^[4]。他们的模型使用了 XGBoost 和 OPTUNA 技术。他们还应用了数据预处理和超参数调优技术。

Mohammed Marouane Saim 和 Hassan Ammor 比较了三种机器学习算法（SVM、逻辑回归和 KNN）来预测心血管疾病。在他们的项目中，支持向量机的准确率为 86.64%。

3 方法论

在本研究中，我们主要采用三种分类器，它们是 CatBoost、LightGBM 和 XGBoost。对于每个分类器，我们采用以下流程：首先准备数据并进行数据预处理。对于预处理后的数据，我们使用两种方法，一种是使用分类器的默认超参数，另一种是调整超参数。在超参数调整的方法中，第一步是设定超参数学习的目标并进行超参数学习，然后使用学到的超参数对数据进行分类。最后，对比使用超参数调优和使用默认超参数的分类器性能指标。图 1 展示了超参数优化的工作流程。本节包括数据集描述、数据预处理技术、分类算法和超参数调优技术。

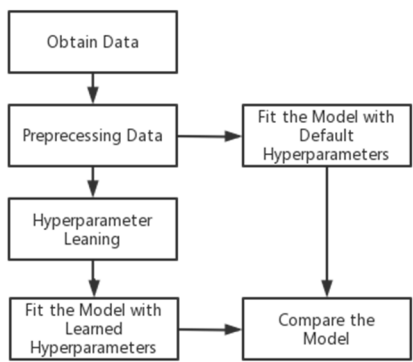


图1. 方法论的工作流程

3.1 数据集

该数据集包含12列，HeartDisease是输出变量，而包括年龄、性别、胸痛类型、静息血压、胆固醇、空腹血糖、静息心电图、最大心率、运动诱发心绞痛、Oldpeak和ST_Slope在内的11列是输入变量。该数据库总共有918条记录，没有缺失值，因此无需处理缺失值。HeartDisease的比例为55.34%，共508名客户；未患心脏病的客户比例为44.66%，共410名客户。从HeartDisease的比例可以看出，数据没有失衡问题。该数据库包含7列数值型数据和5列类别型数据。表1给出了数值型列的统计信息。

表 1. 数值列的统计数据

	Age	休息BP	胆固醇	空腹血糖	最大心率	心电图ST段下移
计算	918	918	918	918	918	918
mean	53.5	132.3	198.7	0.2	136.8	0.8
std	9.4	18.5	109.3	0.4	25.4	1.0
min	28.0	0.0	0.0	0.0	60.0	-2.6
max	77.0	200.0	603.0	1.0	202.0	6.2

Classifiers usually require a large number of data features to train. To identify the most important features that contribute to classifying the result, the author uses Sharpley value as an aid to analyze heart disease. To classify the heart disease data, the author uses random forest classifier with 20 decision trees^[2]. Pal et.al also used Random Forest model to predict heart disease. They obtained 86.9% accuracy with 90.6% sensitivity value and 82.7% specificity value^[3].

Polipireddy Srinivas and Rahul Katarya proposed a novel model hyOPTXg to identify cardiovascular disease^[4]. Their model utilized the XGBoost and OPTUNA technique. They also applied data preprocessing and hyper-parameter tuning techniques.

Mohammed Marouane Saim and Hassan Ammor compared three machine learning algorithms(SVM,Logistic Regression and KNN)to predict cardiovascular diseases^[5]. In their project the accuracy of Suppoert Vector Machine is 86.64%.

3 METHODOLOGY

In this study, we mainly adopt three classifiers, they are CatBoost, LightGBM and XGBoost. For each classifier, we adopt the following process: firstly prepare the data and perform data preprocessing. For the preprocessed data, we use two methods, one is to use the default hyperparameter for the classifier, and the other is to adjust the hyperparameter. Using the method of hyperparameter adjustment, the first step is to set the goal of hyperparameter learning and perform hyperparameter learning, and then classify the data with the learned hyperparameters. Finally, compare the performance metrics of the classifier using hyperparameter tuning and using the default hyperparameters. Figure 1. shows the workflow of hyperparameter optimization. This section includes the dataset description, data preprocessing techniques, classification algorithms and hyperparameter tuning techniques.

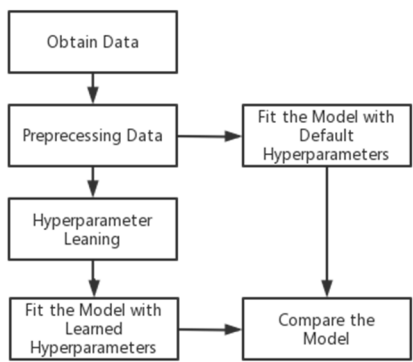


Figure1. Workflow of the methodology

3.1 The dataset

This dataset contains 12 columns, The HeartDisease is the output variable, while eleven input ones including age, sex,ChesPainType,RestingBP, Cholesterol,FastingBS,RestingECG,MaxHR,ExercideAngina,Oldpeak,ST_Slope are input variables. The database contains a total of 918 records with no missing values, so we dot need to deal with missing values. Percentage of HeartDisease is 55.34%, total of 508 customers, percentage of customer did not heartDisease is 44.66%, total of 410 customers. It can be seen from the proportion of HeartDisease that there is no imbalance in the data. The database contains 7 numerical columns and 5 categorical columns. Table 1. gives the statistics of numerical columns.

TABLE 1. THE STATISTICS OF NUMERICAL COLUMNS

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak
count	918	918	918	918	918	918
mean	53.5	132.3	198.7	0.2	136.8	0.8
std	9.4	18.5	109.3	0.4	25.4	1.0
min	28.0	0.0	0.0	0.0	60.0	-2.6
max	77.0	200.0	603.0	1.0	202.0	6.2

3.2 数据预处理

在数据预处理阶段，我们将5个分类变量映射为数值类型。至于胸痛类型，根据其统计特征，我们将ASY映射为1，NAP映射为2，ATA映射为3，TA映射为4。

对于异常值检测，我们定义了一个函数来绘制数值特征的箱线图和分布图。从图中可以看出，RestingBP、Cholesterol 和 Oldpeak 这三个属性中存在异常值。我们删除了对应的记录。图 2 给出了数值特征的箱线图和分布图。

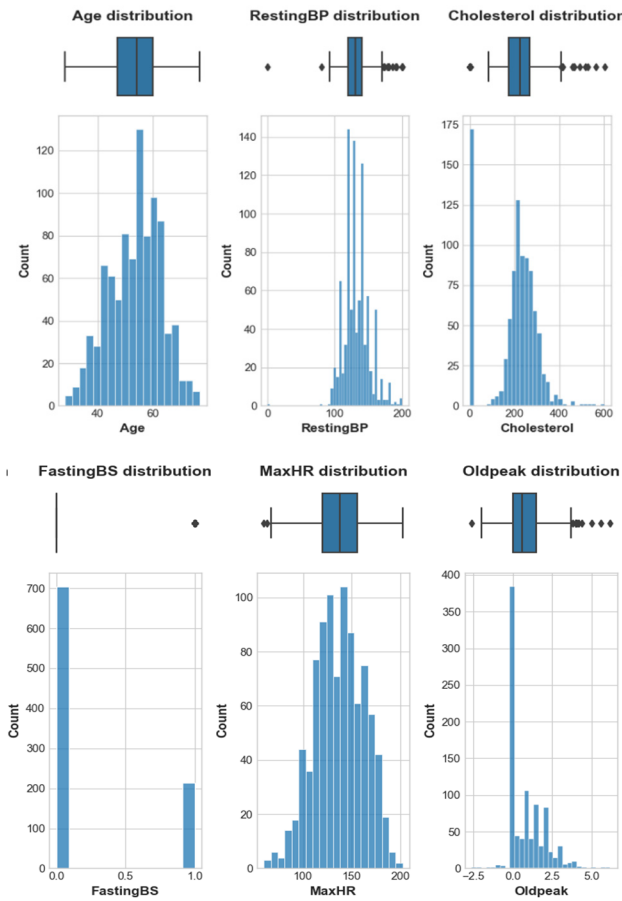


图 2. 数值特征的箱线图和分布。

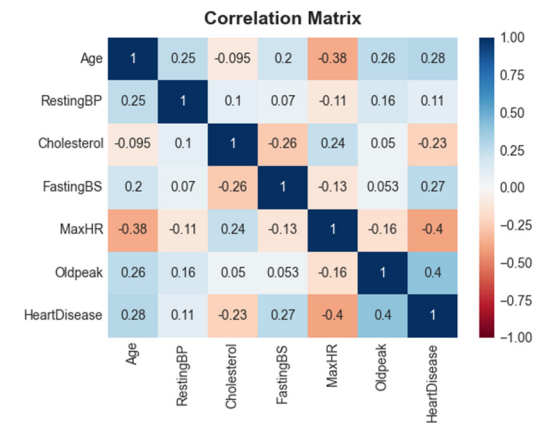


图3. 数值特征的相关性

3.2 Data Preprocessing

In the data preprocessing stage, we map the 5 categorical variables into numeric types. As for ChestPainType, according to its statistical characteristics, we map ASY to 1, NAP to 2, ATA to 3, and TA to 4.

For outlier detection, we define a function to plot both the boxplot and distribution of numerical features. It can be seen from the figure that there are outlier in the three attributes of RestingBP, Cholesterol and Oldpeak. We delete the corresponding records. Figure 2. gives the boxplot and distribution of numerical features.

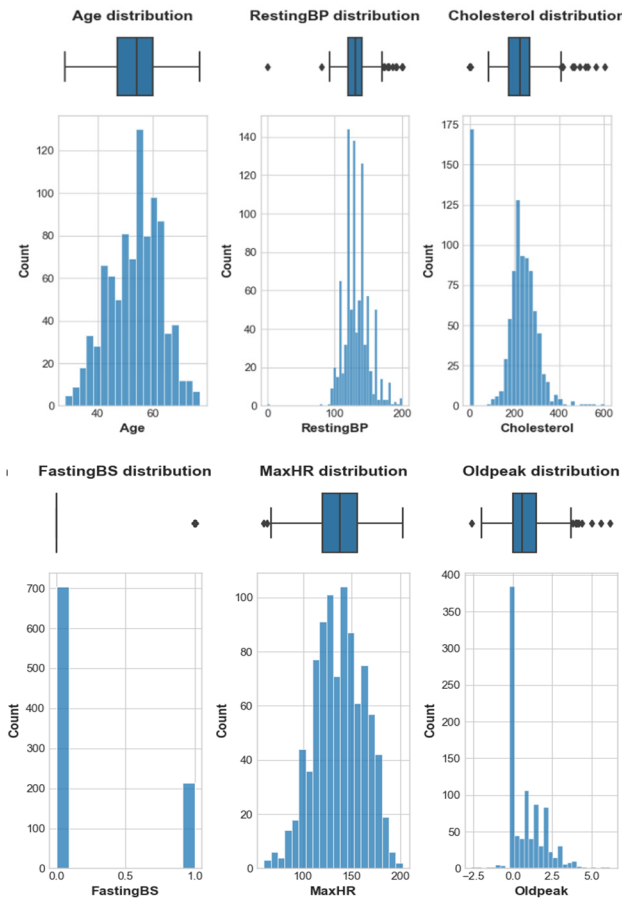


Figure 2. the boxplot and distribution of numerical features.

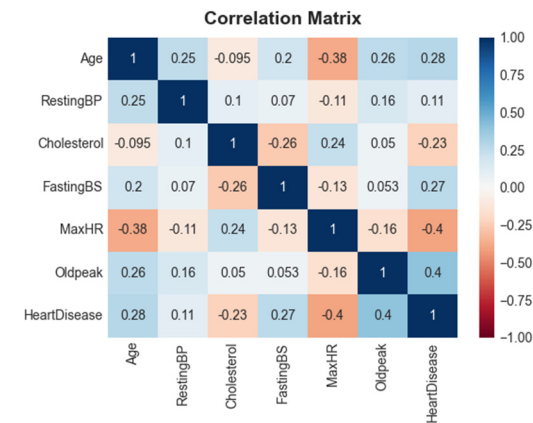


Figure 3. Correlation of numerical features

表 2. 通过贝叶斯优化方法获得的超参数

Name	目标	超参数	超参数重要性
CatBoost	“目标”： "Logloss","colsample_bylevel": 0.01, 0.1, 深度 1, 12, "boosting_type": ["已排序", "Plain"],"引导类型": [“贝叶斯”，“伯努利”，“MVS”]	colsample_bylevel: 0.09932226640716055, 深度: 10, 提升类型: 平淡 引导类型: MVS	colsample_bylevel: 0.77,深度:0.12,提升类型 e: 0.08, 引导类型: 0.03
LightGBM	"目标" 二进制 "二元对数损失","详细程度": -1 "boosting_type": "dart","num_leaves": 2, 2000,"最大深度", 3, 12,"lambda_l1", 1e-8, 10.0, log=True, "lambda_l2", 1e-8, 10.0, 对数=真,"num_leaves", 2, 256,"特征分数", 0.4, 1.0, “bagging_fraction” , 0.4, 1.0, "bagging_freq" , 1, 7, “min_child_samples” , 5, 100	叶子数: 576, 最大深度: 6, lambda_l1: 1.3159635619769443e- 06, lambda_l2: 0.06953260196279283, 特征分数: 0.7930046206900255, 采样比例: 0.9432884010278475, bagging频率: 2, min_child_samples: 22	最小子样本数: 0.72, 特征分数:0.16, 叶子数量: 0.06, 采样比例: 0.03, bagging_freq:0.02, 最大 _深度: 0.01,lambda_l1: 0,lambda_l2:0
XGBoost	冗长 0, “目标”: "binary:logistic","tree_method": "exact", "booster", ["gbtree", "gblinear", "dart"],"lambda", 1e-8, 1.0, log=真, “alpha” , 1e-8, 1.0, og=真, “子采样” 0.2, 1.0,"colsample_bytree", 0.2, 1.0	助推器: gblinear, lambda: 0.002146449268433026, α: 9.721458639113031e-05, 子样本: 0.5699324732532789, colsample_bytree: 0.9414920603461235,	子样本: 0.62, 树列采样率: 0.28, α: 0.05, 助推: 0.04, α: 0.01

图3显示了数值特征的相关性。根据矩阵，我们可以观察到数值特征与目标变量之间的相关性较弱。Oldpeak 与心脏病呈正相关。最大心率与心脏病呈负相关。胆固醇与心脏病呈负相关。

3.3 机器学习方法

在机器学习领域， Boosting算法适用于具有少量训练样本的结构化数据领域。常用的Boosting算法有XGBoost、LightGBM和CatBoost。

XGBoost，也被称为极端梯度提升树，是提升算法的一种实现。XGBoost 是一个用于大规模并行提升树的工具。它的特点是计算速度快且模型性能良好，可用于分类和回归问题。在选择最优分裂点时，XGBoost 先进行预排序，然后计算所有特征的所有分裂点按这些分裂点分割所有样本的目标函数增益。该过程具有较高的空间和时间复杂度。

LightGBM（Light Gradient Boosting Machine，轻量级梯度提升机）是一个用于实现GBDT算法的框架。它使用直方图算法将遍历样本转换为遍历直方图，从而大大降低了时间复杂度；它采用基于叶子节点（Leaf-wise）算法的增长策略来构建树，这减少了许多不必要的计算；它使用直方图算法将存储的特征值转换为存储的分箱值，并且不需要特征值到样本的索引，从而减少了内存消耗；因此，LightGBM可以使用参数“最大树深”来限制树的深度，避免过拟合。

Catboost 可以自动处理类别特征。它基于类别特征的统计数据，计算类别特征的频率，添加超参数，然后生成新的数值特征。Catboost 通过减少大量超参数调节的需求并降低过拟合的可能性，使其更加稳健。此外，CatBoost 还解决了梯度偏差（Gradient Bias）和预测偏移（Prediction shift）的问题，从而减少过拟合的发生，提高算法的准确性和泛化能力。

3.4 超参数调整

超参数调优技术主要包括网格搜索、随机搜索、进化算法和贝叶斯优化。在本文中，我们使用贝叶斯优化方法^[8]。表2给出了参数目标。

TABLE 2. HYPERPARAMETERS OBTAINED BY THE BAYESIAN OPTIMIZATION METHOD

Name	objective	Hyperparameter	Hyperparameter Importance
CatBoost	"objective": "Logloss","colsample_bylevel": 0.01, 0.1,"depth", 1, 12,"boosting_type": ["Ordered", "Plain"],"bootstrap_type": ["Bayesian", "Bernoulli", "MVS"]	colsample_bylevel: 0.09932226640716055, depth: 10, boosting_type: Plain, bootstrap_type: MVS	colsample_bylevel: 0.77,depth:0.12,boosting_type: e: 0.08, bootstrap_type: 0.03
LightGBM	"objective": "binary","metric": "binary_logloss","verbosity": -1, "boosting_type": "dart","num_leaves", 2, 2000,"max_depth", 3, 12,"lambda_l1", 1e-8, 10.0, log=True,"lambda_l2", 1e-8, 10.0, log=True,"num_leaves", 2, 256,"feature_fraction", 0.4, 1.0,"bagging_fraction", 0.4, 1.0,"bagging_freq", 1, 7,"min_child_samples", 5, 100	num_leaves: 576, max_depth: 6, lambda_l1: 1.3159635619769443e- 06, lambda_l2: 0.06953260196279283, feature_fraction: 0.7930046206900255, bagging_fraction: 0.9432884010278475, bagging_freq: 2, min_child_samples: 22	min_child_samples: 0.72,feature_fraction:0.16, num_leaves: 0.06, bagging_fraction: 0.03,bagging_freq:0.02,max _depth: 0.01,lambda_l1: 0,lambda_l2:0
XGBoost	"verbosity": 0,"objective": "binary:logistic","tree_method": "exact","booster", ["gbtree", "gblinear", "dart"],"lambda", 1e-8, 1.0, log=True, "alpha", 1e-8, 1.0, og=True,"subsample", 0.2, 1.0,"colsample_bytree", 0.2, 1.0	booster: gblinear, lambda: 0.002146449268433026, alpha: 9.721458639113031e-05, subsample: 0.5699324732532789, colsample_bytree: 0.9414920603461235,	subsample: 0.62,colsample_bytree: 0.28,alpha: 0.05,booster:0.04,alpha:0.01

Figure 3 gives the correlation of numerical features. Based on the matrix, we can observe weak level correlation between the numerical features and the target variable. Oldpeak has a positive correlation with the heart disease. Maximum heart rate has negative correlation with the heart disease. Cholesterol has negative correlation with the heart disease.

3.3 Machine learning methods

In the field of machine learning, the Boosting algorithm is suitable for the field of structured data with few training samples. Commonly used Boosting algorithms are XGBoost, LightGBM and CatBoost.

XGBoost, also known as extreme gradient boosting tree, is an implementation of the boosting algorithm. XGBoost is a tool for massively parallel boosting trees. It is characterized by fast calculation speed and good model performance, which can be used in classification and regression problems. In the selection of the optimal split point, XGboost performs pre-sorting first, and then calculates the objective function gain of all samples split by these split points for all split points of all features. This process has great space and time complexity^[5].

LightGBM(Light Gradient Boosting Machine) is a framework for implementing the GBDT algorithm. It uses a histogram algorithm to convert traversal samples into traversal histograms, which greatly reduces the time complexity;It adopts the growth strategy based on the Leaf-wise algorithm to build the tree, which reduces a lot of unnecessary computation; It uses a histogram algorithm to convert stored eigenvalues into stored bin values, and does not require the index of eigenvalues to samples, reducing memory consumption; Therefore, LightGBM can use the parameter "maximum tree depth" to limit the depth of the tree and avoid overfitting^[6].

Catboost can automatically handle categorical features. Based on the statistics of categorical features, calculate the frequency of categorical features, add hyperparameters, and then generate new numerical features. Catboost makes it robust by reducing the need for many hyperparameter tuning and reducing the chance of overfitting. In addition, CatBoost also solves the problem of gradient bias (Gradient Bias) and prediction shift (Prediction shift), thereby reducing the occurrence of overfitting, thereby improving the accuracy and generalization ability of the algorithm^[7].

3.4 Hyperparameter tuning

Hyperparameter tuning techniques mainly include Grid Search, Random Search, Evolutionary Algorithms and Bayesian Optimization. In this paper, we use Bayesian Optimization methods^[8]. Table 2. gives the parameter objectives for

优化以及通过贝叶斯优化方法获得的超参数。表格中的第二列是学习目标，第三列是学习到的超参数值，第四列是学习到的超参数重要性。

3.5 结果

在本文中，我们使用准确率、召回率、ROC-AUC 和精确率作为评估我们方法性能的指标。

准确率是指正确预测样本数量与总观测数量的比率，它不考虑预测样本是正样本还是负样本。

Accuracy=(TP+TN)/(TP+TN+FN+FP) (1)

召回率是正确预测为正样本的数量与实际为正类的所有观测值数量的比率。

Recall= TP/(TP+FN) (2)

ROC-AUC 计算 ROC 曲线下的面积以及正例率与假正例率的关系。

TPR = TP/(TP + FN) (3)

FPR = FP/(FP + TN) (4)

精确率被定义为正确预测的正样本数量与所有预测为正样本的数量之比，即所有预测为正的样本中有多少是真正的正样本。

Precision= TP/(TP+FP) (5)

表3展示了三种分类器在有无超参数调优情况下的准确率（Accuracy）、召回率（Recall）、ROC-AUC值以及精确率（Precision）。从表中可以看出，使用超参数调优在准确率、召回率和ROC-AUC指标上都有一定的提升。三种分类器中，使用Optuna进行调优的CatBoost性能最高。

表3. 分类结果

	准确性	召回	ROC_AUC	精度
Catboost	0.8950	0.8990	0.8953	0.8725
Catboost_optuna	0.9041	0.9192	0.9054	0.8750
LightGBM	0.8804	0.8902	0.8782	0.9068
LightGBM_optuna	0.8877	0.9024	0.8843	0.9080
XGBoost	0.8767	0.8788	0.8769	0.8529
XGBoost_optuna	0.8858	0.9091	0.8879	0.8491

我们还将我们的方法与分类方法进行比较，例如逻辑回归、支持向量机、随机森林模型、朴素贝叶斯分类器和Anirban Ghosh 使用的线性判别分析。结果表明，使用这三种 GDBT 模型在准确率、召回率和精确率方面表现更为突出。图 4 给出了不同分类器的性能比较。

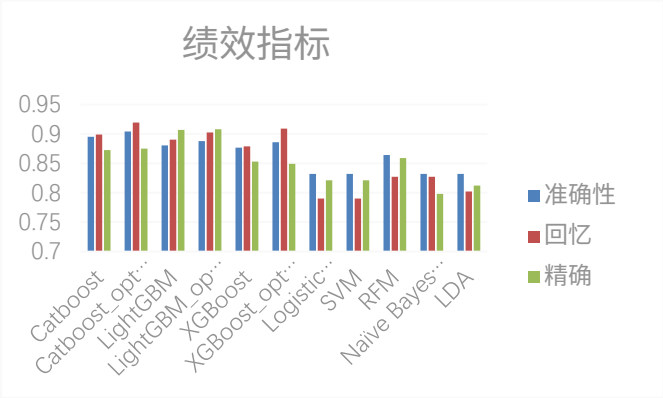


图4. 不同分类器的性能比较 自豪

optimization and the hyperparameters obtained by the Bayesian Optimization method. The second column in the table is the learning objective, the third column is the learned hyperparameter values, the fourth column is the learned hyperparameter importance.

3.5 Result

In this paper, we use accuracy, recall, ROC-AUC, and precision as metrics to evaluate the performance of our methods.

Accuracy refers to the ratio of the number of correctly predicted samples to the total observations, and it does not consider whether the predicted samples are positive or negative.

Accuracy=(TP+TN)/(TP+TN+FN+FP) (1)

Recall is the ratio of the number of correctly predicted positive samples to the all observations in actual class - yes.

Recall= TP/(TP+FN) (2)

ROC-AUC computes the area under the ROC curve and is defined as the false-positive rate versus the true positive rate.

TPR = TP/(TP + FN) (3)

FPR = FP/(FP + TN) (4)

Precision is defined as the ratio of correctly predicted positive obserbations divided by the number of all predicted positive samples, that is, how many of all predicted positive samples are true positive samples

Precision= TP/(TP+FP) (5)

Table 3. presents the values of Accuracy, Recall, Roc-Auc, and Precision for the three classifiers with and without hyperparameter tuning. As can be seen from the table, using hyperparameter tuning has a certain amount of improvement in Accuracy, Recall, Roc-Auc metrics. Three classifiers Catboost with optuna has the highest performance.

TABLE 3. CLASSIFICATION RESULTS

	Accuracy	Recall	Roc_Auc	Precision
Catboost	0.8950	0.8990	0.8953	0.8725
Catboost_optuna	0.9041	0.9192	0.9054	0.8750
LightGBM	0.8804	0.8902	0.8782	0.9068
LightGBM_optuna	0.8877	0.9024	0.8843	0.9080
XGBoost	0.8767	0.8788	0.8769	0.8529
XGBoost_optuna	0.8858	0.9091	0.8879	0.8491

We also compare our method with classification methods such as logistic regression, support vector machine, random forest model, naive bayes classifier and linear discriminant analysis used by Anirban Ghosh^[1].The results show that using these three GDBT models has a more prominent performance on Accuracy, Recall, and Precision. Figure 4.gives performance comparison of different classifiers.

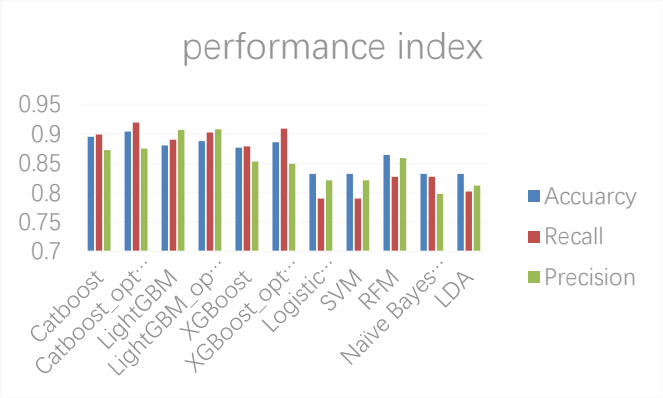


Figure 4.Performance comparison of different classifiers

4 结论

在本研究中，为了找到使用GBDT模型预测心脏病的最佳模型，我们比较了三种GBDT模型，即XGBoost、LightGBM和CatBoost，用于预测一个人是否患有心脏病。我们通过调整超参数来比较这些模型，以提高心脏病预测的准确性。为了比较结果，我们创建了三个使用默认超参数的基础模型。研究发现，使用CatBoost的超参数调优方法进行心脏病预测可以达到平均90%的准确率。结果还表明，调整超参数可以提高模型的预测效果。

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4 CONCLUSION

In this study, in order to find a best model using GBDT models to predict heart disease we compare threeGBDTmodels i.e., XGBoost, LightGBM and CatBoost to predict whether a person has heart disease or not. We compare the models tuning the hyperparameter to improve heart disease predicting. To compare results, we create three base models with default hyperparameter. The study found that using the CatBoost hyperparameter tuning method for heart disease prediction can achieve an average accuracy of 90%. Results also indicated that tuning the hyperparameters improved modeling results.

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